

Unit – IV

Dr. P Arunkumar

Association Rule Mining

Uncovering Hidden Patterns in Data

What is Association Rule Mining?

- A rule-based machine learning method to find interesting "if-then" relationships in large datasets.
- The most famous application is known as "**Market Basket Analysis.**"
- Its goal is to discover strong rules that identify patterns of co-occurrence.
- It helps answer: "What items are often purchased together?"
- The classic example: {Diapers} $\rightarrow$ {Beer}



The Building Blocks



Itemset

A collection of one or more items. (e.g.,
{Milk, Bread, Eggs})



Association Rule

An implication of the form $X \rightarrow Y$, where
 X (antecedent) and Y (consequent) are
itemsets.



Transaction

A single set of items that appear
together in one event (e.g., a single
purchase).

Key Metrics (1/3): Support

Popularity

How often does an itemset appear?

What is Support?

Support measures the frequency of an itemset in the dataset. It's the fraction of total transactions that contain that itemset.

Key Metrics (2/3): Confidence

Likelihood

How strong is the "if-then" rule?

What is Confidence?

Confidence measures the probability that item Y will be purchased when item X is purchased. It measures the strength of the rule.

Key Metrics (3/3): Lift

What is Lift?

Lift measures how much more likely Y is purchased when X is purchased, while controlling for how popular Y is. It shows the *true* strength and value of the association.

$$\text{Lift} (X \rightarrow Y) = \frac{\text{Confidence} (X \rightarrow Y)}{\text{Support} (Y)}$$

Interpreting Lift

- **Lift > 1:** Positive correlation (items are likely bought together). This is a strong, useful rule.
- **Lift = 1:** No correlation (items are independent). The rule is not interesting.
- **Lift < 1:** Negative correlation (items are substitutes). One item's presence discourages the purchase of the other.

The Apriori Algorithm

-  **Set Thresholds:** First, define a **minimum support** and **minimum confidence** value.
-  **Find Frequent Itemsets:** Identify all itemsets in the data that meet the minimum support threshold.
-  **Use the Apriori Principle:** "If an itemset is infrequent, all of its supersets are also infrequent." This **prunes** the search and makes the algorithm efficient.
-  **Generate Rules:** From the list of frequent itemsets, generate all rules that meet the minimum confidence threshold.

A Simple Example: Applying Apriori

1. Sample Transactions

TID	Items
T1	{Milk, Bread, Diapers}
T2	{Milk, Beer, Bread}
T3	{Diapers, Beer}
T4	{Milk, Bread, Beer}
T5	{Diapers, Milk, Bread}

Thresholds:

Minimum Support = 60% (must be in 3/5 transactions)

Minimum Confidence = 80%

2. Finding Frequent Itemsets

Frequent 1-Itemsets (Support ≥ 3):

{Milk}: 4 (80%)

{Bread}: 4 (80%)

{Diapers}: 3 (60%)

{Beer}: 3 (60%)

Frequent 2-Itemsets (Support ≥ 3):

{Milk, Bread}: 4 (80%)

(Others like {Milk, Diapers}: 2, {Milk, Beer}: 2, {Diapers, Beer}: 1 are pruned)

Frequent 3-Itemsets:

(No 3-itemsets can be formed from only {Milk, Bread}. Search is pruned.)

3. Generating Rules

From {Milk, Bread}:

Rule: {Milk} $\rightarrow$ {Bread}

Confidence = $4 / 4 = 100\%$. (Meets 80% min) $\rightarrow$ **Strong Rule!**

Real-World Applications

- **Retail:** Product placement in stores and cross-selling ("Customers who bought this also bought...").
- **E-commerce:** Recommendation engines for movies (Netflix), music (Spotify), or products (Amazon).
- **Healthcare:** Discovering co-occurrence of symptoms and medical conditions or predicting adverse drug reactions.
- **Fraud Detection:** Identifying unusual patterns or combinations of financial transactions.

